

Medical artificial intelligence for clinicians: the lost cognitive perspective



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The development and commercialisation of medical decision systems based on artificial intelligence (AI) far outpaces our understanding of their value for clinicians. Although applicable across many forms of medicine, we focus on characterising the diagnostic decisions of radiologists through the concept of ecologically bounded reasoning, review the differences between clinician decision making and medical AI model decision making, and reveal how these differences pose fundamental challenges for integrating AI into radiology. We argue that clinicians are contextually motivated, mentally resourceful decision makers, whereas AI models are contextually stripped, correlational decision makers, and discuss misconceptions about clinician–AI interaction stemming from this misalignment of capabilities. We outline how future research on clinician–AI interaction could better address the cognitive considerations of decision making and be used to enhance the safety and usability of AI models in high-risk medical decision-making contexts.

Introduction

Medical decision systems based on artificial intelligence (AI) are used in many radiology departments, with more than 350 systems approved by the US Federal Drug Administration.¹ However, the development and commercialisation rate of AI decision systems far outpaces our understanding of their value for clinicians,² with the gap between technological advancement and clinical value being described as an AI chasm.³ Technical and logistical problems, and challenges for clinical implementation, are potential contributors to this AI chasm.³ Although technology implementation in other high-risk settings, such as increased automation in aeroplane cockpits, has been previously investigated to understand and improve use,^{4,5} evaluating AI implementation for clinicians remains a neglected area of research on medical AI.²

The human–machine relationship describes the complex and sometimes illogical outcomes of human decision making alongside technology, such as over-reliance, under-reliance, and overwhelm.^{5,6} At present, the underexplored relationship between clinicians and AI relies too heavily on self-reported, surface-level analysis.^{7,8} To better address some of these challenges to human implementation and ultimately bridge the AI chasm, we draw on previously unapplied perspectives from the research areas of cognitive science and human factors for a theoretical and empirical understanding of human decision making in clinical contexts. We describe key distinctions between clinicians and AI, highlighting the consequent knowledge gaps and misconceptions surrounding their combination. We use radiology as a case study domain of expertise as most current AI models are used for diagnostic decisions from medical imaging data and it is the most well studied area of medical cognition. However, we expect these principles to apply to all forms of medical reasoning, and hence use terms radiologist and clinician interchangeably. We hope this approach allows for better navigation of AI use for clinicians and directions for AI developers to allow this technology to achieve its full potential.

Clinical reasoning is based on cue utilisation

Radiologists diagnose by noticing cues from the clinical environment.^{9,10} Cues are sensory signals that guide clinician attention and prompt the use of abstract knowledge representations called inner cognitive cues.¹¹ The clinical environment is rich with sensory cues that radiologists use to diagnose, even if they are unnoticeable to the novice observer. For example, nodule brightness on a mammogram could indicate the presence of a specific type of tumour, or specific symptoms listed on the imaging request form could affect how sensitive a radiologist will be to finding features.¹²

With experience, radiologists learn which cues guide their attention towards the most clinically relevant information in their environment.^{12,13} This ability to use domain-relevant information is known as cue utilisation¹¹ and it is a hallmark of expertise. This expertise enables radiologists to rapidly extract the gist or essential features from the clinical scene while remaining highly accurate, guiding subsequent processing and analysis of specific clinical features.¹⁴ Other similar processes that involve quick cue utilisation include the concept of the clinical gestalt, where specific patterns of signs and symptoms intuitively signal a medical emergency or abnormality.¹⁵ Indeed, methods of decision making that compress and make use of the surrounding context to simplify the task at hand are just as effective as more complicated methods—a principle known as ecological rationality in cognitive science.¹⁶

Reasoning is bounded

Rationality is known in cognitive science as a type of formal reasoning that is a hypothetically optimal approach to making decisions.¹⁶ Historically, presumptions about so-called good decision making relied on a view of traditional rationality, where ideal human reasoning is expected to conform to abstract mathematical rules of logic and probability.¹⁶ For instance, clinicians are commonly encouraged to follow Bayes' rule when diagnosing patients,¹⁷ in which the posterior probability of a disease is estimated from previous factors and test sensitivity and specificity.¹⁷

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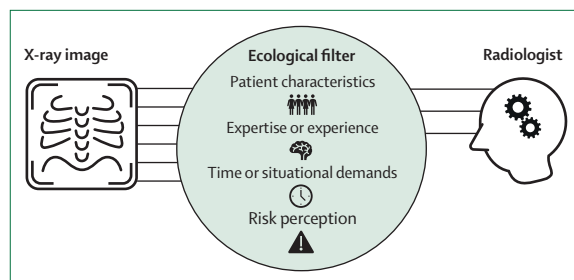


Figure 1: Representation of the ecological filter radiologists use to examine clinical information

This filter allows most visual information from the medical image to be readily discarded leaving the most salient, important, and clinically useful cues.^{12,16,18}

However, humans are limited by sensory, cognitive, and time constraints.¹⁶ Thus, rather than optimising when reasoning (ie, striving for the single best possible decision given perfect and complete information), humans actually satisfice, a process in which they make a decision that is good enough for the task based on only the available information.¹⁶ Satisficing without making worse decisions is possible because rationality is dependent on the environment, which dictates what decisions are rational or good.

For instance, in a clinical setting familiar to the clinician, most possible decisions are easily discarded; an otherwise healthy patient with a sore leg after exercise is unlikely to have an advanced malignancy and the clinician can ignore this possibility entirely, unless relevant evidence to the contrary presents itself. In this way, the environment and its cues reduce the number of possible decisions by placing bounds on what is probable and, in doing so, allow the clinicians to rapidly consider a smaller range of potential decisions (figure 1).¹⁶ Hence, humans can make efficient decisions based on context instead of complete knowledge of all possible options and outcomes.¹⁶

By being ecologically rational, clinical context guides clinicians on what cues to look for, which is why expert decision makers are known to use less information than non-experts—ie, use of the gist signal.^{19,20} Targeted information use is important because of humans limited processing capacity to integrate all information sources into decisions,¹⁹ unlike new deep learning systems that are far less inhibited by information processing constraints than humans and allow for large amounts of information to be incorporated into every decision. However, AI systems also do not have an explicit ability to ignore extraneous cues to simplify their decisions, making them fundamentally different decision makers to clinicians. An AI model cannot question its dataset in the same way clinicians are encouraged to question the validity of what they have been taught: a practice in the clinical setting called epistemic humility.^{21,22} Although we recognise this is a simplification of clinician and AI decision making, this distinction in capabilities is almost certainly valid across health care and AI development currently.

Furthermore, as we draw distinctions between clinicians and AI models that use rationality, we recognise that humans are imperfect and retain some unjustified judgments based on race, gender, etc. We therefore focus on how the ecologically rational model of decision making has a tendency towards non-spurious associations as an important difference between AI and clinicians. This mismatch in decision making is largely unexplored but suggests that the integration of AI into clinician decision making is far more complex than currently acknowledged.²³

AI models rebound clinical context

In keeping with the ecologically bounded model of cognition (which sees rational and optimal decision making as rooted in environmentally valid inferences), bounded rationality describes a clinician's ability to be accurate by being frugal with the information they use. We introduce the notion of rebounding to refer to the opposite of this process, in which decisions once rooted in a few valid external cues are instead made through all available information, even if this is not cognitively optimum, comprehensible, or visible to the clinician. For example, AI models can use image cues that were useful in the context of the training dataset but are undeniably unrelated to the clinical task, such as the presence of radiographical tokens (ie, written markers outside the image) to predict whether a patient has pneumonia.²⁴

This use of extraneous information can be understood as the rebounding of the task from the clinical context and occurs in two stages: labelling and modelling. Human input is used during labelling, in which experts distil their decision making into a simple, often binary indicator of the presence or absence of pathology, and in doing so discard the vast majority of the clinical context that informed the decision in the process.^{20,25} Salient information about which cues are important is lost, and all that remains is the final expert decision itself (ie, the label). From here, human input ends and transitions to the modelling phase, in which the AI system is provided a combination of inputs (eg, medical images) and labels that describe the expert decision and is then directed to learn a mapping from one to the other—to predict a label from an image (eg, produce a diagnosis). The modelling stage can be seen as the inverse of labelling and the goal is to reconstruct the decision process with perfect fidelity. This second phase can also be considered a form of targeted filtering, where the AI learns to discard most unimportant information in the input images by selecting image features that are most useful for the task.^{24,26,27}

Features are learnt through trial and error, a strict mathematical optimisation process based on previous output error (known as the loss function) that usually minimises the overall error rate.²⁶ However, this process consists of largely unguided optimisation—AI learns any pattern useful for the task of predicting labels from inputs given the specific training dataset, regardless of whether those environmental cues are ecologically rational or

would be readily discarded by a human expert.^{24,26,28–30} Thus, models find and incorporate any feature that reduces loss and correlates with diagnosis, which debounds images from their ecological features that informed the original label (figure 2).^{24,28–30}

This debounding from ecological features can allow models to make use of irrational features that translate poorly outside of test conditions, known as shortcut learning—eg, use of radiographical markers, artifacts of hospital processes, and biologically unrelated features of the patient as cues to diagnose pneumonia.^{24,28–30} This two-step AI development process produces a mismatch between AI and human decision makers, in effect highlighting the distinction between ecological rationality and traditional rationality. However, AI is not perfectly traditionally rational as it exists in the context of imperfect information;³¹ models themselves are only as good as their training data.^{32,33}

With this understanding, AI models produce decisions that are no longer exclusively based on ecologically valid inferences. Rather, these decisions become diluted in an unknown pool of cues learned from the training data, which the model has found mathematically useful to improve its accuracy but that might not be clinically valid. Rather than clinically useful bounds, decisions by AI models are dataset bounded: they use whatever features in the training data correlate with the labels. Research suggests that as AI models become more accurate, they become less aligned with human decision-making processes.³⁴ This mismatch has subsequent implications on black box problems and model explainability (see the section on explainability).³⁵

Bounding mismatch: implications for decision making

The complexity of the clinician–AI interaction lies in the combination of these two separate entities that are mismatched in their capabilities or boundedness. AI systems present decisions to clinicians that follow different logic to humans, leaving clinicians unable to easily understand what information has been lost or incorporated in that decision. Obtaining this understanding is nearly impossible and the cues used by AI systems are often unable to be fully determined.³⁵ This mismatch introduces vulnerabilities to clinical decision making, such as limited ability to anticipate AI errors or recognise AI biases.³⁶ Clinicians, without complete understanding of the AI decision-making process, must face a difficult choice between avoiding both over-reliance, in which they uncritically accept AI output and its potential pitfalls, and under-reliance, in which AI output is avoided altogether.^{37,38} The following section outlines how current approaches to clinician–AI interaction overlook this bounding mismatch.

Teaming

Research in this area often attempts to fuse AI and clinicians together through the idea of a clinician–AI team³⁹

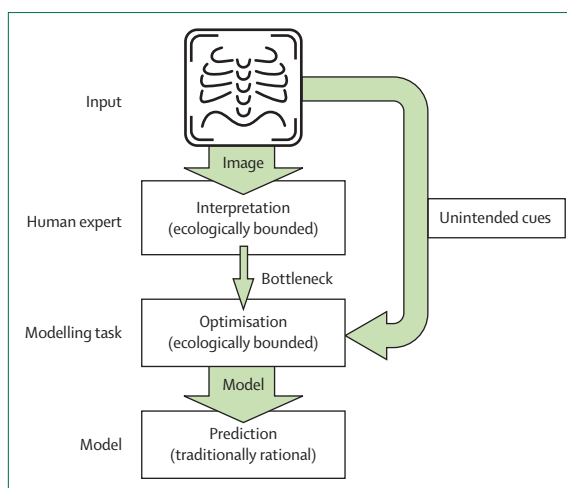


Figure 2: Representation of the debounding of clinical decision making through deep learning models

Inputs such as x-ray images are distilled into a label by a human expert by making use only of relevant image cues, based on ecological and contextual relevance. The label is passed to the AI model, which takes the input image and learns to recognise image characteristics that are mathematically correlated with the label but might not be ecologically relevant to the task. Hence, the AI diagnosis becomes unbounded, meaning unintended or irrelevant cues from the images can be used for decision making. The difference in arrow size from the interpretation box to the prediction box represents the bottleneck of information involved in the debounding process, and the arrow to the right represents the unintended and uncontrolled introduction by the model of features or predictors that human experts do not find useful for the clinical task.

through human-like communication to achieve the trust and understanding seen in human teams.^{40,41} However, the mismatch between these decision makers can be understood as a disruption of the typical function of human teams. By sharing the same mechanisms that make decisions, humans can reliably judge others' decisions on the basis of communication and explanation.^{42,43} Therefore, trust between humans is a naturally built result of this shared understanding.⁴³ However, AI–clinician teams do not have this shared understanding, most clearly shown by the field of explainable AI (XAI).

Explainability

The field of XAI, broadly spanning technical AI development to implementation, seeks to provide some explanation of the AI process to a human user.^{40,44} In the past few years, the rapid rate of AI implementation in high-risk settings has led to policy makers calling for transparent and explainable algorithm–decision processes to increase human comprehension.⁴⁵ Although it is important that AI is understandable to users, explainability as a focus for improving decision making with AI is flawed when considering that AI and clinician decision making are affected by mismatched goals (ie, mathematical optimisation vs clinical care) and logic (ie, correlational reasoning vs ecological reasoning).^{16,20,28–30} Therefore, attempts at explanation fundamentally fail on the basis of little common understanding.

Whereas human teams can rely on each other's explanations due to shared human capabilities and cognitive mechanisms, AI possesses different and often incomprehensible capabilities.^{35,46} Without shared capabilities underpinning the interaction between humans and AI, explainability cannot replicate the reliable ecological qualities of human-to-human communication.^{42,46} Despite this difference, AI is often discussed and understood by users in highly anthropomorphic or human-like ways, in which systems appear misleadingly capable of communicating, thinking, and emoting like humans.⁴⁷ These misconceptions contradict the belief that medical AI systems are tools that should only facilitate decisions rather than replace them.^{2,23–25} XAI has thus been linked to an illusion of explanatory depth, whereby XAI can cause user understanding of AI to be inaccurate and overconfident due to misunderstanding AI capabilities.⁴⁶

The value of XAI in improving decision making is highly inconsistent, distanced from theoretical cognitive understanding, and associated with worsened AI-assisted decisions in some experiments.^{46,48–50} Even from a technical perspective, AI explanations in clinical systems only offer an approximation of the algorithmic processes that led to the output, presenting just a proportion of the complexity underlying explanation.^{35,51} We argue that research on AI use by clinicians must move beyond observable shared behaviours into an understanding of cognition in the context of AI. We suggest that the three levels of behavioural analysis proposed by Marr⁵² can similarly be applied to the study of human–AI interaction (figure 3).

First, the behavioural level is the outermost view of human–AI interaction and studies their observable behaviour as a team with a collective goal. Observed outcomes include trust, certainty, task performance, and explainability. Although useful, this level provides little insight beyond observation and cannot connect to the cognitive processes that produced specific outcomes. Even focusing on cognitive biases, such as automation bias (defined as a bias towards misusing technological

aids through over-reliance, under-reliance, or reduced vigilance for errors), merely describes rather than explain cognition.⁵³ Second, the cognitive level goes deeper into the human mental mechanisms by isolating semi-observable cognitive processes underlying decisions, such as attention, memory, cognitive load, and calibration. These processes are only somewhat observable; they can be inferred through carefully designed experiments grounded in theory that isolate and examine these processes in different contexts. An example might be an experiment in which participants complete a task with AI assistance and attentional load is manipulated between groups to investigate whether AI use is affected by increased attentional load. Third, the cognitive model level complements the cognitive level by studying these processes in their interconnected form through mathematical models. These models are developed and tested to account for biological and environmental constraints on cognitive processes, allowing a more comprehensive understanding of human cognition.⁵⁴ The models allow an understanding of cognition in its complexity, and they are re-evaluated with new knowledge about neurophysiological constraints. For instance, cognitive attention during a task can be studied within a model that also accounts for changing decision thresholds and response time.⁵⁵ An example for medical AI might be a cognitive model that investigates how the previous probabilities doctors have on the incidence of specific pathologies, which are known to affect decision thresholds, can affect doctors' attentional capacity when using AI on more difficult pathology cases. This model would allow investigation of how AI could modify those earlier distributions of pathology prevalence and therefore clinicians' attention to some cases, which could be used to improve AI use within the context of their expertise.

Expanding research into levels two and three will explore how observable and latent factors underlying clinical decision making change and adapt in the context of AI use. This research will allow recognition of how clinical decisions themselves also change and adapt in response to clinical experience, environmental pressures,

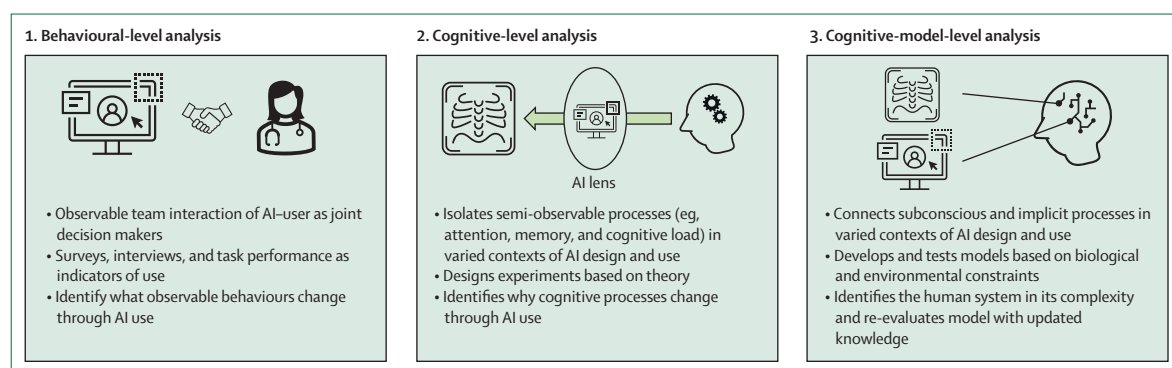


Figure 3: Three levels of analysis—behavioural, cognitive, and cognitive model
Adapted from Marr.⁵² AI=artificial intelligence.

patient interaction, and outcomes. A cognitive approach has been crucial when exploring the effects of automated decision aids on human users before the advent of modern AI—eg, autopilots in aviation.^{6,56} For instance, the introduction of GPS navigation, terrain displays, and further technological aids to the aircraft cockpit has notoriously led to increased distraction and difficulties for pilots.^{4,6,56} These effects have been explored behaviourally (through observational studies of pilot trust and cognitive load);⁵⁷ cognitively (through studies of attention in varying conditions);⁵⁸ and in modelled form (developing a mechanistic understanding, such as the drift diffusion model).⁵⁹ However, these earlier technologies were far simpler to understand, less autonomous and anthropomorphised, and less central to core expert tasks than current AI.⁶ In legal and policy-making settings, liability surrounding the interaction between clinician and AI is gaining substantial interest.⁶⁰ Thus, a comprehensive understanding from the cognitive sciences is crucial to develop new theory for AI-assisted medical decisions. Such understanding will be important to develop standards and regulations to ensure system safety in contexts in which future instances of misuse are inevitable and liability is unclear.⁶⁰ AI has great potential in medicine, and by incorporating the considerations outlined in this Viewpoint, we can better ensure it benefits clinicians and patients.

Conclusions and takeaways for future research

Although it is common to refer to clinicians and AI as a synergistic team, this view is misleading. Human teams rely on a shared understanding of their tasks and shared cognitive mechanisms between members. However, human–AI teams are mismatched: clinicians are ecologically bounded, with their decisions constrained by their knowledge and environment, whereas AI is dataset bounded, as it learns any useful correlation in its training data and is not limited by context. Therefore, a comprehensive understanding of clinical cognition in the context of AI use is necessary for the safe development and implementation of medical AI. With deep-learning based technologies becoming far more autonomous and removed from the scope of human understanding, while also being misleadingly portrayed as human-like interfaces, there is an important need to change how clinician cognition is approached in the context of AI use. This change requires the following: understanding that cognitive decision making is a crucial part of both model development (ie, image labelling) and model use (ie, clinician interpretation); emphasising the fundamental differences between humans and AI when evaluating their respective decision making; and acknowledging the internal complexity of decision making by expanding research focus away from purely observational approaches and exploring the cognitive, environmental, and neurophysiological aspects of human decision making in the context of AI use.

Contributors

LT: conceptualisation, literature review, theoretical framework, diagrams and figures, collaboration and discussion, writing—original draft, and writing—review and editing. CS: conceptualisation, literature review, theoretical framework, diagrams and figures, collaboration and discussion, and writing—review and editing. MM: collaboration and discussion and writing—review and editing. RS: collaboration and discussion and writing—review and editing. MG: collaboration and discussion and writing—review and editing. LO-R: conceptualisation, literature review, theoretical framework, diagrams and figures, collaboration and discussion and writing—review and editing.

Declaration of interests

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